

Methanolysis of polyethylene terephthalate (PET) in the presence of aluminium triisopropoxide catalyst to form dimethyl terephthalate and ethylene glycol

Hideki Kurokawa*, Masa-aki Ohshima, Kazuo Sugiyama, Hiroshi Miura

Department of Applied Chemistry, Faculty of Engineering, Saitama University, Shimo-Okubo 255, Saitama-shi, Saitama 338-8570, Japan

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Abstract

Aluminium triisopropoxide (AIP) promoted the methanolysis of polyethylene terephthalate (PET) to form monomers, dimethyl terephthalate (DMT) and ethylene glycol (EG), in an equimolar ratio. The methanolysis at 200 °C in methanol with an AIP catalyst gave DMT and EG in 64% and 63% yields, respectively. The yields were increased by using a toluene/methanol mixed solvent containing 20–50 vol.% toluene; maximum yields, 88% for DMT and 87% for EG, were obtained at 20 vol.% toluene. These results indicate that the rate of methanolysis strongly depends on the solubility of PET. The results of GPC analysis suggest that the methanolysis of PET in the absence of the catalyst includes three steps. In the first step, the depolymerisation occurred at a tie molecule connecting PET crystals and the chain length was shortened to about 1/3. The shortened chain was depolymerized to oligomers in the second step. The GPC curve of the oligomers tailed to low molecular weight, clearly indicating that the depolymerization took place at random positions on the polymer chain. The third step, the depolymerisation from the oligomers to the monomers, was promoted only in the presence of the AIP catalyst.

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1. Introduction

Consumption of polyethylene terephthalate (PET) has increased considerably in recent years, because it has come into use for beverage containers. Used PET containers are mainly recycled to polyester textiles (material recycle), but the supply of PET for recycling will exceed the demand for recycled textile for the foreseeable future. Chemical recycling of PET to monomers constitutes effective resource utilization, because processes for such recycling can be carried out repeatedly without substantial waste. Three methods (hydrolysis, glycolysis and methanolysis) have been proposed for the chemical recycling of PET [1]. In hydrolysis, PET is depolymerized to terephthalic acid (TPA) and ethylene glycol (EG) with water, when necessary, under acidic or basic conditions [2–6]. Alkali

decomposition of PET in non-aqueous media to form TPA salt, which is analogous to hydrolysis under basic conditions, has also been reported [7–8]. Meanwhile, glycolysis of PET with ethylene glycol in the presence of a catalyst yields bis-2-hydroxyethyl terephthalate (BHET) [9–12].

Methanolysis of PET affords dimethyl terephthalate (DMT) and ethylene glycol. These products are recycled to the transesterification process of PET production and enjoy some advantages over the products of hydrolysis and glycolysis; for example, easy purification steps. Methanolysis has been attempted by Lotz et al. [13] and Gruschke et al. [14], as described in the references, which are patents. In the process of Gruschke et al., PET was completely (more than 99%) depolymerized to DMT and EG by reacting molten PET with methanol at 210 °C in the absence of a catalyst. Some patents [15–18] teach improved processes for increasing the quality of the products and lowering recycling costs.

Sako et al. [19,20] proposed a new process of methanolysis by treating PET with supercritical methanol. In

* Corresponding author. Tel.: +81-48-858-9547; fax: +81-48-858-9547.

E-mail address: kuro@apc.saitama-u.ac.jp (H. Kurokawa).

this process PET can be completely depolymerized to DMT, EG, and oligomers above 300 °C at 11 MPa for 30 min without a catalyst.

As is well known, methanolysis, which is a form of transesterification, proceeds effectively in the presence of catalysts. The patent literature discloses useful catalysts for the methanolysis of PET, such as zinc acetate, manganese salts (acetate and phosphate), calcium phosphate, oxides and hydroxides of alkali metals, oxides and hydroxides of alkali-earth metals, sodium silicate, and lead oxide [2–5], but the results of reactions using such catalysts are not actually shown. Meanwhile, results are shown for use of magnesium acetate for glycolysis and use of stannous oxalate for alcoholysis with an alcohol having 6–15 carbon atoms [12,21].

Although chemical recycling of PET by methanolysis requires catalysts exhibiting high activity, the depolymerisation of PET in the presence of catalysts has not been widely studied. In the present work, we have investigated the methanolysis of PET to form DMT and EG in the presence of aluminium triisopropoxide (AIP). AIP serving as a Lewis acid catalyst is known to promote transesterification of dialkylcarbonates with phenol [22].

2. Experimental

Aluminium triisopropoxide (>95%) was purchased from Wako Pure Chemical Industries, Ltd. Methanol and toluene were supplied from Kanto Chemical CO., Inc. Those chemicals were used without further purification. PET specimens were prepared by cutting a beverage bottle to square chips measuring 3 mm×3 mm. The catalyst, the PET chips, and the solvent were charged into an autoclave equipped with a magnetic stirrer and a purge valve. After being sealed, the autoclave was heated, while the solution was stirred, to reaction temperature (160–200 °C) by use of an oil bath. Methanolysis was carried out for 2 h, after which the reactor was immediately cooled in an ice bath. The reaction mixture was dissolved in 100 ml methanol. About 1 ml of the suspended solution was filtered, and the filtrate was analysed by a gas chromatograph equipped with a capillary column (DB-WAX, 30 m). Finally, the remaining solution was filtered, and the residue was dried and weighed as a recovered PET.

The average molecular weight and molecular weight distribution of the PET specimen were determined by a GPC (TOSOH, SEC-PAD) equipped with an UV detector, and a TSKgel GMHhr-M column (length = 60 cm). The values were normalized by use of polystyrene standards. A scanning electron microscope (SEM, Hitachi S-4100) was used for surface observation of the recovered PET specimen.

3. Results and discussion

3.1. Methanolysis in methanol

Detailed results are summarized in Table 1. The first run (Run 1) was a blank test carried out at 200 °C in the absence of a catalyst. Small amounts of DMT and EG were formed, in yields of 2.9% and 3.6%, respectively. After the products were dissolved in 100ml of methanol, most of the PET (92.2%) was recovered as chips by filtration, and a large number of checkered cracks [Fig. 1 (a) and (b)] were observed on the chip surface by SEM. These results indicate that most of the PET chips failed to dissolve in methanol under the reaction conditions.

Runs 2–5 show the results of methanolysis at 160–200 °C in the presence of AIP catalyst. At 160 °C, the methanolysis proceeded only slightly (Run 2), and PET was recovered in the same shape as before [Fig. 1(c)]. At 180–190 °C, PET was depolymerized to DMT and EG (10–20% yield) in equimolar ratio (Run 3–4). The form of recovered PET was the same as before, but the surface of PET chips became rough [Fig. 1(d)]. These results suggest that PET was gradually depolymerized to the monomers from the chip surface. When the temperature was raised from 190 to 200 °C, the yields of DMT and EG increased dramatically, from 18.6% and 18.7% (Run 4) to 67.3% and 68.3% (Run 5), respectively. The insoluble component was recovered as powdery PET. About 27% of the charged PET could not be recovered after methanolysis at 200 °C, indicating that the unidentified products dissolved in methanol at ambient temperature. Methanolysis of PET is generally considered to take place at random positions on the polymer chain, decreasing the chain length and finally producing DMT and EG. In view of both solubility in methanol and the above reaction path, the unrecovered component is assumed to exist as oligomers. Indeed, Sako et al. determined the structure of oligomers in methanolysis products by field desorption mass spec-

Table 1
PET methanolysis in methanol in the presence of aluminium triisopropoxide^a

Run no.	Reaction temp (°C)	Catalyst charged		Yield (%)		Recovered PET ^b	
		(mg)	(nmol)	DMT	EG	(%)	Shape
1	200	0	0	2.9	3.6	92.2	Chip + powder
2	160	50	0.245	0.8	0	100	Chip
3	180	50	0.245	10.5	8.8	84.1	Chip
4	190	50	0.245	18.6	18.7	76.0	Chip
5	200	50	0.245	67.3	68.3	4.6	Powder
6	200	32	0.157	57.4	60.9	12.3	Powder
7	200	12	0.059	20.2	24.0	60.8	Chip

^a Reaction conditions; reaction time = 2 h, PET = 0.50 g (2.60 mmol unit), methanol = 10 ml (247 mmol).

^b Recovered from the reaction mixture by filtration.

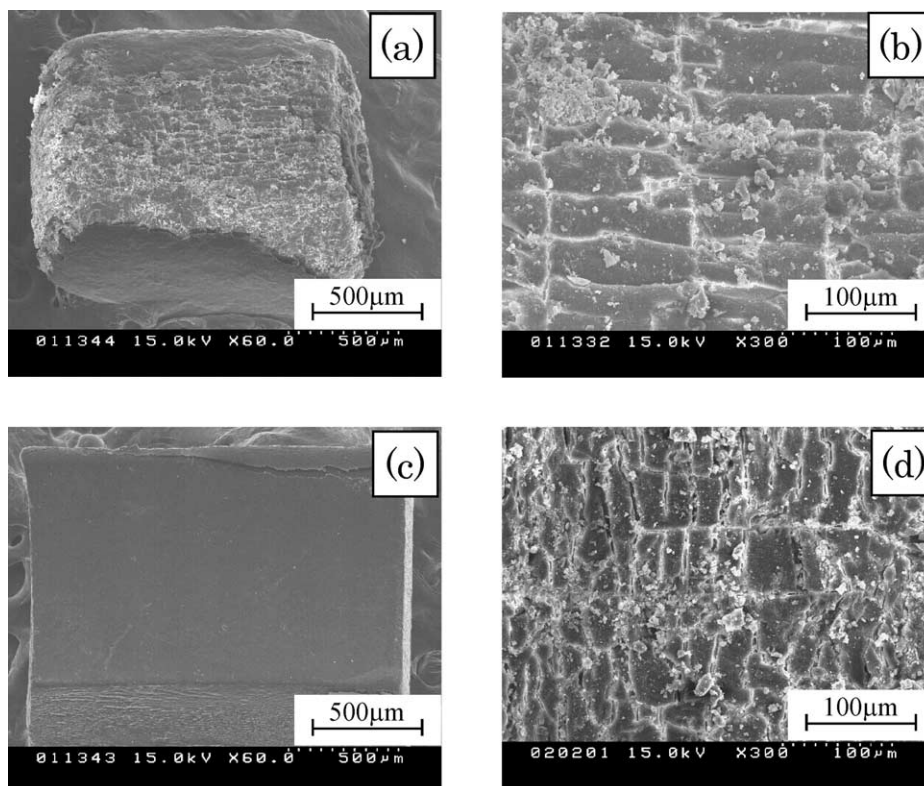


Fig. 1. SEM images of PET recovered after methanolysis. The samples were obtained by methanolysis (a), (b) at 200 °C in the absence of catalyst, (c) at 160 °C in the presence of catalyst, and (d) at 190 °C in the presence of catalyst.

trometry; the structures were mainly $\text{HOOC-C}_6\text{H}_4\text{-COOCH}_2\text{CH}_2\text{OH}$ and $\text{HOOC-C}_6\text{H}_4\text{-COOCH}_2\text{-CH}_2\text{OOC-C}_6\text{H}_4\text{-COOH}$ [19].

Runs 5–7 represent the relationship between yield and the amount of catalyst charged. The yields of monomers increased with the amount of catalyst up to 32 mg and did not depend on the catalyst loading at yields higher than 60% because of equilibrium between the monomers and oligomers.

3.2. Methanolysis in methanol/toluene mixed solvent

The AIP catalyst promotes the methanolysis of PET in methanol at 200 °C, however, the yields of DMT and EG are insufficient for practical use. The moderate rate of depolymerization is attributed to the low solubility of PET in methanol. A modification with mixed solvent was previously carried out by Hu et al. in the methanolysis of polycarbonate to form dimethyl carbonate and bisphenol A [23]. They reported that methanolysis was promoted in methanol/toluene mixed solvent containing 20 vol.% toluene.

To increase the solubility of PET, a mixed solvent of methanol and toluene was employed. The results of PET methanolysis in methanol/toluene are summarized in Table 2. The yield of DMT increased from 67.3% in methanol (Run 5) to 82–89% in the mixed solvent containing 20–50 vol.% toluene (Runs 9–11). Powdery

PET was recovered after the blank test (Run 8) in 20 vol.%-toluene/methanol, whereas original PET was recovered after methanolysis in methanol without AIP catalyst (Run 1). These results indicate that the solubility of PET increased by addition of toluene to methanol.

When toluene content was raised from 50 vol.% to 80 vol.%, the yields of DMT and EG decreased from 82.0% and 83.4% (Run 11) to 25.2% and 31.6% (Run 13), respectively. One possibility is the lower concentration of methanol.

Table 2
PET methanolysis in methanol/toluene in the presence of aluminium triisopropoxide^a

Run no.	MeOH:Tol (ml:ml)	Yield (%)		Recovered PET ^b	
		DMT	EG	(%)	Shape
8 ^c	(8:2)	2.9	4.6	67.9	Powder
9	(8:2)	88.5	87.2	0	—
10	(6:4)	86.8	85.1	0	—
11	(5:5)	82.0	83.4	0	—
12	(4:5)	63.6	62.3	12.4	Powder
13	(2:8)	25.2	31.6	62.0	Powder

^a Reaction conditions: reaction temperature = 200 °C, reaction time = 2 h, PET = 0.50 g (2.60 mmol unit), catalyst weight = 50 mg (0.245 mmol), solvent = 10 ml.

^b Recovered from the reaction mixture by filtration.

^c In the absence of catalyst.

Methanolysis of PET under severe conditions is known to yield diethylene glycol (DEG) [1]. This condensation of EG is catalyzed by an acidic-COOH group of partially hydrolysed PET [24]. The formation of DEG was also reported in hydrolysis of PET in the presence of an acid catalyst [25]. Therefore, we carefully analyzed the formation of DEG by a GC in Run 8. As a result, we confirmed the formation of DEG in small quantity (about 0.8%), but did not detect 2-methoxyethanol formed by condensation of ethylene glycol with methanol. These results clearly indicate that the AIP catalyst does not promote the undesirable consecutive reaction of EG even when EG concentration is relatively high.

3.3. GPC analysis

The average molecular weights (M_n and M_w) of recovered PET were determined by GPC for investigating the details of methanolysis, and Fig. 2 shows the GPC curves. Average molecular weight of the starting PET was $M_n=16,000$, and its distribution was $M_w/M_n=3.4$ [Fig. 2 (a)]. After methanolysis at 200 °C in the absence of catalyst (Run 1), PET was recovered as chips (Fig. 2 (b)) with a small amount of powder. M_n of the powder portion (4300) was nearly equal to that of the chips (5500) and about one-third lower than that of the starting PET. The molecular weight distributions of

both powder and chips became sharp in comparison with that of the initial PET. These findings suggest that, when M_n is above 4000, depolymerization does not take place at random positions of the polymer chain. When methanolysis was carried out in the presence of the catalyst (Run 6, AIP = 32 mg), most PET was depolymerized to the monomers and 12.3% for the initial charge was recovered. M_n of the recovered PET was equal to 2200 and its GPC curve tailed to low molecular weight [Fig. 2 (c), $M_w/M_n=2.6$]. This means the occurrence of depolymerization at random positions on the polymer chain.

As shown in Fig. 2 (d), the molecular weight distribution of the recovered PET in Run 8 (methanol/toluene mixed solvent, no catalyst) is broad ($M_w/M_n=3.8$), and peaks derived from oligomers are observed in the low molecular weight region, clearly indicating that PET was easily depolymerized in the mixed solvent at random positions on the chain. M_n (1400) of recovered PET in Run 8 was one-quarter that (5500) in Run 1 and half that in Run 6, but the yield (2.9%) of DMT in Run 8 is extremely low in comparison with that (57.4%) in Run 6.

The results of GPC analysis suggest that the methanolysis in the absence of a catalyst includes at least three steps represented in Fig. 3. In the first step of methanolysis, the chain length of PET was shortened to about 1/3 [Fig. 3 (i)]. Even when the catalyst is not used, the first step takes place easily [Fig. 2 (b)]. The details of this step are not well understood, however the polymer chain might be cut off at a tie molecule because such molecules exist in an amorphous part of PET and have high accessibility. Collins and Zeronian [26] reported similar results for alkaline hydrolysis of fabric PET; after the hydrolysis, crystalline material was deposited on the fibre surface, and its molecular weight was approximately 2400.

A significant difference in GPC curves was observed between samples obtained by reacting PET in methanol and those in toluene/methanol mixed solvent. In toluene/methanol, most PET was depolymerized to oligomers in the absence of the catalyst [second step, Fig. 3 (ii)] but the peak ($\log M=4$) derived from the crystalline material was still shown in Fig. 2 (d). This fact denotes that the oligomers are produced through the first step. The mixed solvent accelerates the formation of oligomers effectively and this promoting effect is due to swelling of PET crystals.

The third step [Fig. 3 (iii)], the formation of monomers from oligomers, is promoted only in the presence of catalyst and the monomers were produced in high yield. Those results suggest that the catalyst promotes the formation of monomers from oligomers.

In conclusion, aluminium triisopropoxide promotes the methanolysis of PET to form the monomers DMT and EG. The yields of monomers strongly depend on

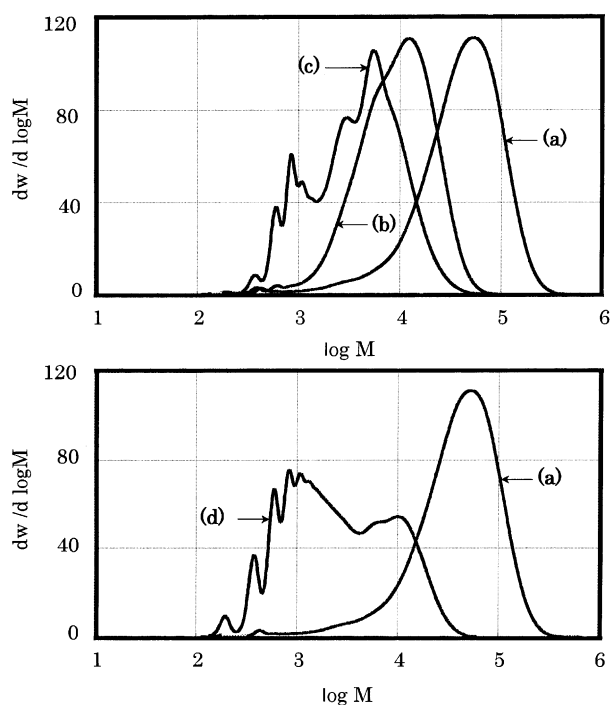


Fig. 2. GPC curves of PET recovered after methanolysis. Before methanolysis. After methanolysis in methanol (b) at 200 °C in the absence of catalyst, and (c) at 200 °C in the presence of catalyst. (d) After methanolysis in methanol/toluene mixed solvent in the absence of catalyst.

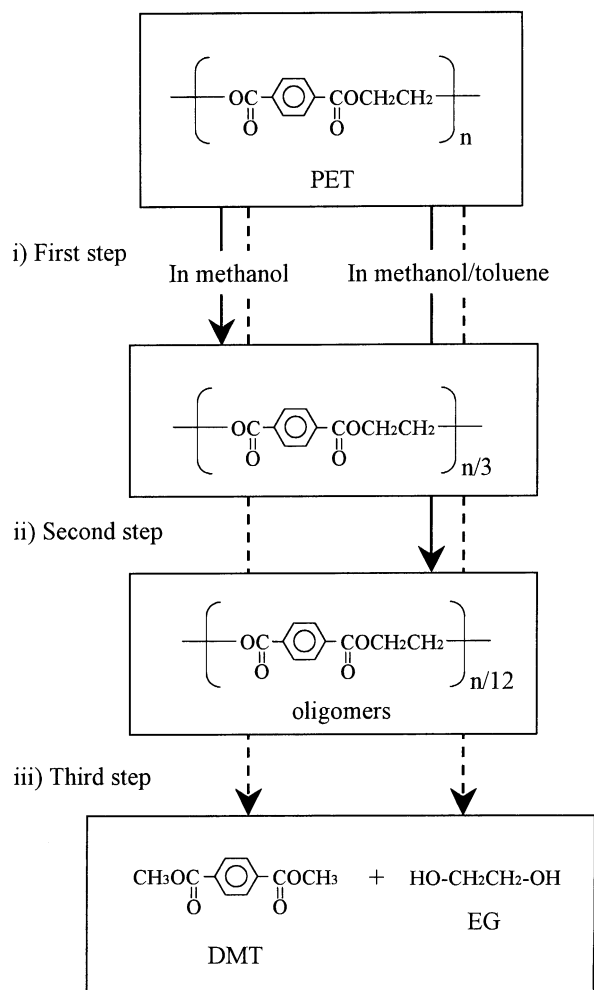


Fig. 3. Reaction scheme of PET methanolysis. Solid line: without catalyst; broken line: with catalyst.

the solubility of PET; methanolysis in 20 vol.-%-toluene/methanol at 200 °C in the presence of an aluminium triisopropoxide catalyst produced DMT in 88.5% yield and EG in 87.2% yield.

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